



Department of Physics

National Institute of Technology Kurukshetra

Course: B.Tech (EE, SET, CE, ECE, ME, PIE, R&A) Exam: Sessional-I Date: 19-09-2024
Subject: Engineering Physics Course Code: PHIC 101 Duration: 50 min

Note: All questions are compulsory.		Max. Marks: 20	Roll No: 124111039	
1.(a) Why is population inversion essential for LASER operation, what conditions are necessary to achieve it, and what methods are used to accomplish it?				3+2
(b) If the refractive index of the LASER medium is 1.5, calculate the wavelength of the emitted LASER light, which has a wavelength of 600 nm in vacuum, when it propagates within the medium.				
2. For a step-index optical fiber with core and cladding refractive indices of 1.54 and 1.5, respectively, calculate the following parameters when the fiber is placed in a medium with a refractive index of 1.45: (a) Critical angle, (b) Fractional refractive index, (c) Numerical aperture, and (d) Acceptance angle.				1+1+1+2
3.(a) Was the Michelson-Morley experiment a failure or a success? Provide justification for your answer. (b) If the Ether as a medium doesn't exist, how does sunlight travel to the Earth? (c) Write down Inverse Lorentz Transformation Equations for space and time.				2+1+2
4.(a) How does the interaction of positrons with matter differ from that of electrons? (b) Which particles—electrons, protons, or neutrons—participate in nuclear reactions? Justify your answer.				3+2

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